Hojoon Kim

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Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Republic of Korea
B.S. in Computer Science, Minor in Electrical Engineering — *Mar. 2021 – Present*

- **GPA**: 4.14/4.30

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
Exchange Student in Computer Science — *Feb. 2024 – Jul.2024*

- **Coursework:** Advanced Compiler Construction, Intro to Machine Learning, Computer Vision, Undergrad Research

Research Interest

My research interests lie at the intersection of system architecture and complex challenges. I am passionate about applying core principles of computer systems to optimize high-level problems in areas like computer vision.

Research Experience

Computer Architecture and Systems Lab (CASYS) — Daejeon, Republic of Korea
Research Intern; *Advisor: Jongse Park* — *Sep. 2025 – Present*

- Investigating 3DGRUT performance on GPUs to explore acceleration opportunities at both HW and SW levels.
- Developed foundational knowledge in graphics (shaders, shading languages, graphics APIs) and ray tracing.
- Set up reproducible Docker environments for 3DGS, Vulkan-Sim, and 3DGRUT-on-Vulkan-Sim.
- Analyzed Vulkan-Sim and GPGPU-Sim traces to characterize 3DGRUT execution on simulated GPUs.

Parallel Systems Architecture Lab (PARSA) — Lausanne, Switzerland
Visiting Researcher; *Advisor: Babak Falsafi* — *Mar. 2024 – Jun. 2024*

- Contributed to the Midgard project, a novel virtual memory design that divides address translation into two parts.
- Proposed a lazy invalidation technique for translations in the Midgard virtual memory system, reducing unnecessary cache flushes and enhancing memory management efficiency.

Computer Architecture and Systems Lab (CASYS) — Daejeon, Republic of Korea
Undergraduate Researcher; *Advisor: Jongse Park* — *Jun. 2023 – Jan. 2024*

- Contributed to a Video Understanding project that exploited inter-frame redundancy in ViT
- Analyzed ViT acceleration potential by computing FLOPs ratios across Video-Language Models.
- Conducted comprehensive literature reviews, summarizing relevant research papers for the project.

Industry Experience

RTst — Daejeon, Republic of Korea
RTOS Researcher — *Oct. 2024 – Present*

- Contributed to developing **RTWORKS**, the Real-Time Operating System for reliable and trustworthy systems.
- Designed and implemented kernel functionalities and device drivers, including software timers and the SATA driver.
- Ported RTWORKS to several target boards, developing board support packages (BSPs) for each platform.

Publications

Déjà Vu: Efficient Video-Language Query Engine with Learning-based Inter-Frame Computation Reuse
Jinwoo Hwang, Daeun Kim, Sangyeop Lee, Yoonsung Kim, Guseul Heo, **Hojoon Kim**, Yunseok Jeong, Tadiwos Meaza, Eunhyeok Park, Jeongseob Ahn, Jongse Park
VLDB 2025. [paper]

Teaching/Mentoring

Teaching Assistant: KAIST CS101 Introduction to Programming — *Sep. 2023 – Dec. 2023*
Mentor: KAIST CS101 Introduction to Programming — *Sep. 2023 – Dec. 2023*

Services

Team Leader: of Director at KAIST Broadcasting System
Association Member: at KAIST Freshman Student Council & KAIST School of Computing Student Council